

Saaransh Gupta

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EDUCATION

Thapar Institute of Engineering and Technology

Bachelor of Engineering in Computer Sciences

Patiala, Punjab

2022 – 2026

BCM Arya International

12th Grade (A levels)

Ludhiana, Punjab

2020 – 2022

EXPERIENCE

Full Stack Intern

Apr 2024 – Feb 2025

ACT Center, Thapar School of Liberal Art and Sciences

Patiala, Punjab

- Developed a **NextJS** application to parse **BibTeX** files, dynamically displaying publications on **TIET's official website**.
- Built a **full-stack** web application using **React**, **NodeJS** and **PostgreSQL** to visualize GitHub commit activity, enabling real-time tracking of junior developers' contributions and improving team workflow efficiency by 30%.
- Implemented a **RESTful API** using **NodeJS** to enable efficient data retrieval for a Learning Management System.

Full Stack Intern

Aug 2024 – Dec 2024

PraxisForge Pvt. Ltd.

Remote | Delhi, India

- Contributed over **10,000** lines of code to the company's repository fixing bugs across domains increasing efficiency by 20%.
- Pioneered an application to simplify the **editorial workflow** by enabling review of news articles prior to publication.
- Integrated databases with a Flutter application, leveraging **MongoDB** and **Redis** for caching frequently accessed data.

Undergraduate Research Intern

Sep 2022 – Sep 2023

Pollution Control Board (PPCB), Ministry of Environment

Patiala | Chandigarh, India

- Contributed to an **Industry Institute Interface**, strengthening link between **ITIs** and industry partners across the state.
- Managed daily logs and automated communications using existing systems to enhance operational efficiency by over 40%.
- Collaborated with **PPCB officials** and **TIET management** to identify real-world problems and optimize existing solutions.

PROJECTS

BoxBox - Formula 1 Strategy Maker | Python, NumPy, Pandas, React

Feb 2025 – Mar 2025

- Conceived a tyre degradation model using **regression** to predict tyre usage across laps and different rubber materials.
- Programmed a race strategy simulator leveraging data from **FastF1 API** for prediction of optimal pitstops in a race.
- Optimized pit strategies using statistical modeling, and **Monte Carlo** simulations aimed at race-time minimization.

EasyFind - Smart Lost & Found System | NodeJS, React, Appwrite, Python

Aug 2024 – Oct 2024

- Streamlined Thapar University's lost-and-found item tracking system through **AI-powered** image captioning.
- Improved owner identification, reducing search and wait times impacting over **2,000 users** across the campus.
- Engineered CI/CD workflows with **GitLab** to automate deployment of **AppWrite** backend and **HuggingFace** AI services.
- Fine-tuned **Salesforce's BLIP** model, boosting accuracy to **80%** for precise and accurate **image captioning**.

Splitzee - Split Funds Easily | NodeJS, React, MongoDB

Jun 2024 – Jul 2024

- Built Splitzee, a **full-stack** web application for seamless group expense tracking and bill-splitting among friends.
- Implemented core features such as adding expenses, assigning payers, and auto-splitting bills across selected participants.
- Utilized **TanStack Query** for client-side caching, reducing traffic on backed services and boosting performance by **40%**.
- Integrated social features including friend management and trip-based expense grouping to enhance user collaboration.

ACHIEVEMENTS

- Technical Secretary** at Google Developers Club (**GDC**); led tech initiatives, organized developer workshops across TIET.
- Core member of **Creative Computing Society** (CCS), working on official TIET projects and fostering teamwork skills.
- Represented Ludhiana at State Level Badminton championship in 2020

SKILLS

Languages: JavaScript, Python, GoLang, C/C++, SQL (Postgres, Oracle), HTML/CSS, R

Frameworks: NextJS, React, NodeJS, WordPress, Appwrite, Bootstrap, Material UI, Tailwind CSS

Developer Tools: Git, Docker, Google Cloud Platform, AWS, Azure, Visual Studio, PyCharm

Relevant Coursework: Data Structures(DSA), DBMS, CN, OOPS, OS, AIML, NLP, Generative AI